Lctr 7: Continuous-Time LTI Systems and Properties

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Focus: Sections 2.2 and 2.3 (Pages 97 to 115)

7.1 From Discrete-Time to Continuous-Time: The Big Picture

Why This Matters

In Lecture 6, we learned that for discrete-time LTI systems, the output is the **convolution sum**. Now we extend these ideas to **continuous-time** systems. The math changes from sums to integrals, but the intuition remains the same: **if you know how a system responds to an impulse, you can predict its response to any input!**

Quick comparison:

Concept	Discrete-Time (DT)	Continuous-Time (CT)
Signal decomposition	$\sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$	$\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau$
Convolution	$\sum_{k=-\infty}^{\infty} x[k]h[n-k]$	$\int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$
Impulse function	$\delta[n]$ (unit sample)	$\delta(t)$ (Dirac delta)

Key Insight

The key insight: Sums become integrals, and the index k becomes the continuous variable τ. The concepts are the same—only the notation changes!

7.2 The Continuous-Time Unit Impulse (Dirac Delta)

Before we can do convolution, we need to understand the continuous-time impulse function, denoted $\delta(t)$.

7.2.1 What is $\delta(t)$?

The Dirac delta function $\delta(t)$ is **not** a regular function—it's a mathematical idealization. Think of it as:

- An infinitely tall, infinitely narrow spike at $t = 0$
- The area under the spike equals exactly 1

Formally:

$$\delta(t) = 0 \quad \text{for } t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t)\,dt = 1$$

Intuition: Imagine a pulse that gets narrower and taller while keeping its area equal to 1. The Dirac delta is the "limit" of this process.

7.2.2 The Sifting Property (Most Important!)

The sifting property is the key to understanding convolution:

$$\boxed{\int_{-\infty}^{\infty} x(\tau)\delta(\tau - t_0)\, d\tau = x(t_0)}$$

What this means: The integral "sifts out" (extracts) the value of $x(\tau)$ at the point $\tau = t_0$.

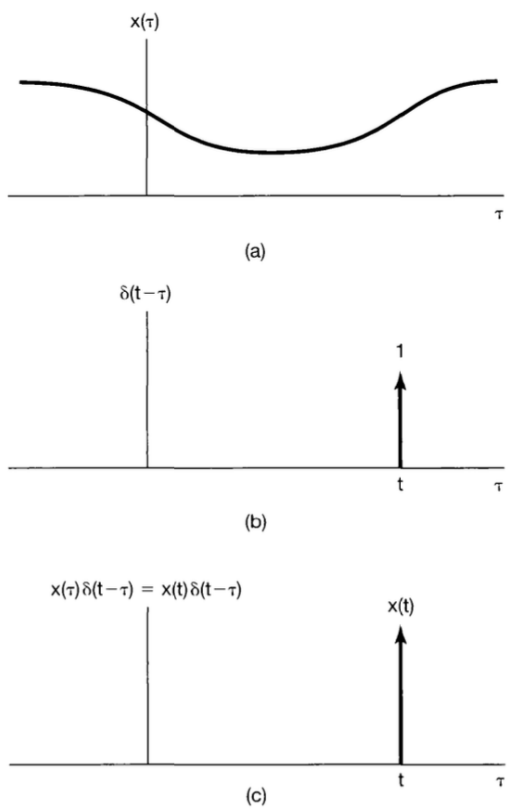


Figure 2.14 (a) Arbitrary signal $x(\tau)$; (b) impulse $\delta(t-\tau)$ as a function of τ with t fixed; (c) product of these two signals.

Figure 1: Figure 2.14: (a) An arbitrary signal $x(\tau)$; (b) the impulse $\delta(t-\tau)$ as a function of τ with t fixed; (c) the product $x(\tau)\delta(t-\tau) = x(t)\delta(t-\tau)$, showing how the impulse "sifts out" the value $x(t)$.

Key Insight

How to use the sifting property:

1. Find where the delta function is "active" (where its argument equals zero)
2. Evaluate the other function at that point
3. That's your answer!

Example 1: Evaluate $\int_{-\infty}^{\infty} (t^2 + 3t)\delta(t-2)\,dt$

Solution:

- The delta is at $t = 2$ (set $t - 2 = 0$)
- Evaluate $(t^2 + 3t)$ at $t = 2$: $(2)^2 + 3(2) = 4 + 6 = 10$
- Answer: $\boxed{10}$

Example 2: Evaluate $\int_{-\infty}^{\infty} e^{-3t}\delta(t+1)\,dt$

Solution:

- The delta is at $t = -1$ (set $t + 1 = 0$)
- Evaluate e^{-3t} at $t = -1$: $e^{-3(-1)} = e^3$
- Answer: $\boxed{e^3}$

Common Mistake

Watch the limits! If the delta function is outside the integration limits, the integral equals zero!

Example: $\int_0^5 x(t)\delta(t+2)\,dt = 0$ because the delta is at $t = -2$, which is outside $[0, 5]$.

7.3 Representing CT Signals Using Impulses

Just like in discrete-time, any continuous-time signal can be written as a "sum" (integral) of shifted, scaled impulses:

$$\boxed{x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau}$$

Interpretation:

- We decompose $x(t)$ into infinitely many infinitesimally small impulses

- Each impulse at time τ has "weight" $x(\tau)\,d\tau$
- The integral adds up all these weighted impulses

This representation is the foundation for deriving the convolution integral.

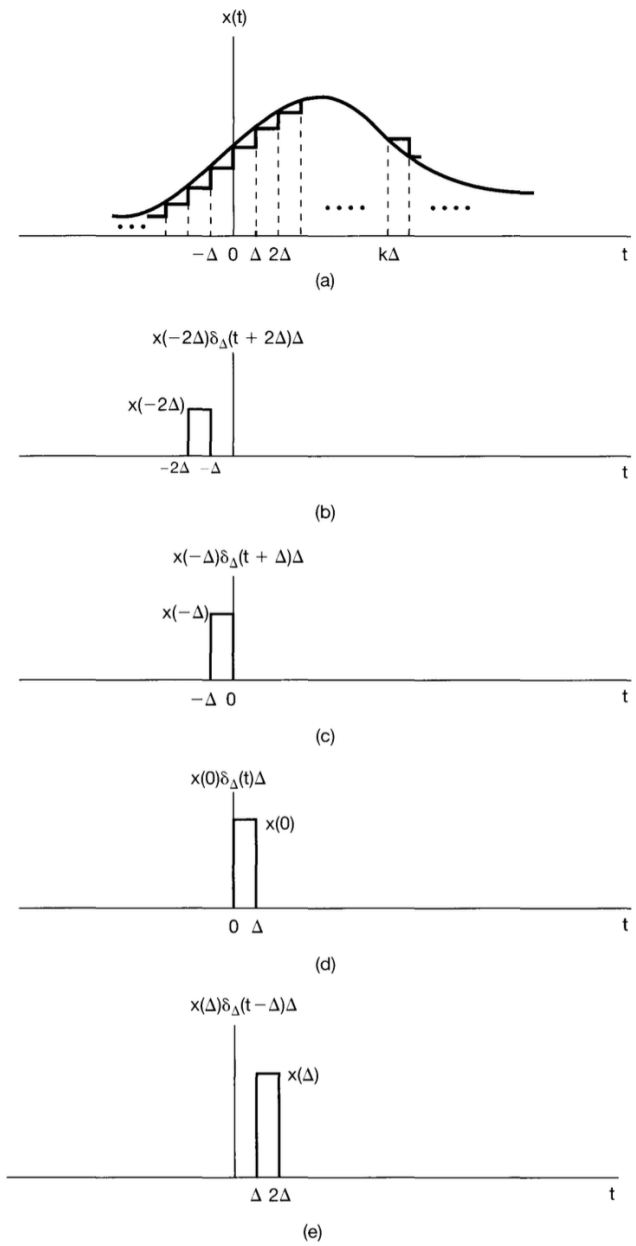


Figure 2.12 Staircase approximation to a continuous-time signal.

Figure 2: Figure 2.12: Staircase approximation to a continuous-time signal. As $\Delta \to 0$, the sum becomes an integral and the pulses become impulses.

7.4 The Continuous-Time Impulse Response

7.4.1 Definition

The **impulse response** $h(t)$ is the output of an LTI system when the input is the unit impulse $\delta(t)$:

$$\boxed{h(t) = T\{\delta(t)\}}$$

Due to time invariance:

- If input is $\delta(t)$, output is $h(t)$
- If input is $\delta(t-\tau)$ (impulse shifted to time τ), output is $h(t-\tau)$

Key Insight

The impulse response completely characterizes an LTI system. Once you know $h(t)$, you can find the output for *any* input!

7.5 The Convolution Integral

7.5.1 Derivation (Step by Step)

This derivation follows the exact same logic as the discrete-time case:

Step 1: Decompose the input

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau$$

Step 2: Apply the system

$$y(t) = T\{x(t)\} = T\left\{\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau\right\}$$

Step 3: Use LINEARITY (move T inside the integral)

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\cdot T\{\delta(t-\tau)\}\,d\tau$$

Step 4: Use TIME-INVARIANCE

$$T\{\delta(t-\tau)\} = h(t-\tau)$$

Step 5: The Convolution Integral

$$\boxed{y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau = x(t) * h(t)}$$

7.5.2 The Flip-and-Shift Interpretation

To compute $y(t)$ for a specific value of t:

1. **Flip:** Take $h(\tau)$ and flip it about $\tau = 0$ to get $h(-\tau)$
2. **Shift:** Shift by t to get $h(t - \tau)$
3. **Multiply:** Multiply $x(\tau)$ by $h(t - \tau)$
4. **Integrate:** Integrate the product over all τ

Common Mistake

Just like in discrete-time, you must **flip before shifting**! The argument $h(t - \tau)$ means: flip h, then shift it by t. Forgetting the flip is the most common mistake!

7.5.3 Alternative Form (Commutativity)

By a change of variables, we can also write:

$$y(t) = \int_{-\infty}^{\infty} x(t - \tau)h(\tau)\, d\tau$$

This shows that convolution is **commutative**: $x(t) * h(t) = h(t) * x(t)$

Key Insight

Practical tip: You can flip either x or h—choose whichever is simpler!

7.5.4 How to Find the Cases and Integration Limits (Step-by-Step Method)

This is one of the most challenging parts of convolution. Here's a systematic approach:

Key Insight

The Golden Rule: The integration limits are determined by the **overlap** between $x(\tau)$ and $h(t - \tau)$. You only integrate where **both** functions are nonzero!

Step-by-Step Procedure:
Step 1: Identify the "support" of each signal
The **support** is the range of values where the signal is nonzero.

- For $x(\tau)$: Find where $x(\tau) \neq 0$. Call this range $[a, b]$.
- For $h(t-\tau)$: Start with where $h(\tau) \neq 0$, say $[c, d]$. After flipping and shifting, $h(t-\tau) \neq 0$ for $t - d \leq \tau \leq t - c$.

Step 2: Find the overlap region
The integrand $x(\tau) \cdot h(t-\tau)$ is nonzero only where **both** signals are nonzero:

$$\text{Overlap} = [a, b] \cap [t-d, t-c]$$

The integration limits are:

$$\text{Lower limit} = \max(a, t-d) \qquad \text{Upper limit} = \min(b, t-c)$$

Step 3: Find the "critical values" of t
The cases change when the overlap relationship changes. This happens when:

- The left edge of $h(t-\tau)$ reaches the left edge of $x(\tau)$: $t-d=a \Rightarrow t=a+d$
- The left edge of $h(t-\tau)$ reaches the right edge of $x(\tau)$: $t-d=b \Rightarrow t=b+d$
- The right edge of $h(t-\tau)$ reaches the left edge of $x(\tau)$: $t-c=a \Rightarrow t=a+c$
- The right edge of $h(t-\tau)$ reaches the right edge of $x(\tau)$: $t-c=b \Rightarrow t=b+c$

Sort these critical values to find the boundaries between cases.
Step 4: For each case, determine the limits
For each range of t values between critical points:

- Lower limit of integral $= \max(\text{left edge of } x, \text{left edge of } h(t-\tau))$
- Upper limit of integral $= \min(\text{right edge of } x, \text{right edge of } h(t-\tau))$

Common Mistake

Common error: Students often set limits incorrectly. Remember:

- If $x(\tau)$ is nonzero on $[a, b]$, these are **fixed** limits (don't depend on t)
- If $h(t-\tau)$ is nonzero on $[t-d, t-c]$, these limits **move with** t

Detailed Example: Finding the Cases
Let $x(\tau) = u(\tau) - u(\tau-1)$ (pulse from 0 to 1) and $h(\tau) = u(\tau) - u(\tau-2)$ (pulse from 0 to 2).
Step 1: Identify supports

- $x(\tau) \neq 0$ for $0 \leq \tau \leq 1$, so $a=0$, $b=1$
- $h(\tau) \neq 0$ for $0 \leq \tau \leq 2$, so $c=0$, $d=2$
- After flip-and-shift: $h(t-\tau) \neq 0$ for $t-2 \leq \tau \leq t$

Step 2: Critical values of t
Calculate when edges align:

- Right edge of $h(t-\tau)$ reaches left edge of x: $t - 0 = 0 \Rightarrow t = 0$
- Right edge of $h(t-\tau)$ reaches right edge of x: $t - 0 = 1 \Rightarrow t = 1$
- Left edge of $h(t-\tau)$ reaches left edge of x: $t - 2 = 0 \Rightarrow t = 2$
- Left edge of $h(t-\tau)$ reaches right edge of x: $t - 2 = 1 \Rightarrow t = 3$

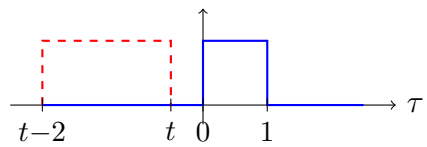
Critical values in order: $t = 0, 1, 2, 3$

Step 3: Analyze each region

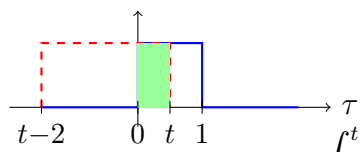
Case	**Range of t**	**Overlap?**	**Integration Limits**
1	$t < 0$	No (h is to the left of x)	None, $y(t) = 0$
2	$0 \leq t < 1$	Partial (h entering)	$\int_0^t$
3	$1 \leq t < 2$	Full overlap with x	$\int_0^1$
4	$2 \leq t < 3$	Partial (h exiting)	$\int_{t-2}^1$
5	$t \geq 3$	No (h is to the right of x)	None, $y(t) = 0$

Graphical View of Each Case:
The following figures show $x(\tau)$ and $h(t-\tau)$ for representative values of t in each case.

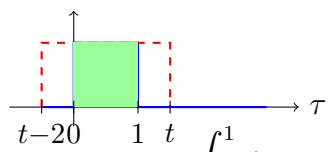
Legend: $x(\tau)$ (fixed) $h(t-\tau)$ (moves right as t increases) (integrate here) Overlap



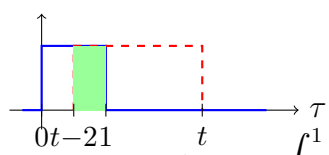
Case 1: $t = -0.5$ (no overlap), $y(t) = 0$



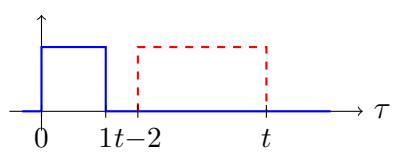
Case 2: $t = 0.5$, limits: $\int_0^t$



Case 3: $t = 1.5$, limits: $\int_0^1$ (full overlap)



Case 4: $t = 2.5$, limits: $\int_{t-2}^1$



Case 5: $t = 3.5$ (no overlap), $y(t) = 0$

Key Insight

Key observation: As t increases, $h(t-\tau)$ slides to the **right**. The green shaded area represents the integral value $y(t)$ at that particular t.

How to read the table:

- **Case 2** ($0 \le t < 1$): The right edge of $h(t-\tau)$ is at $\tau = t$, which is inside $[0, 1]$. The left edge is at $\tau = t - 2 < 0$. So we integrate from $\max(0, t-2) = 0$ to $\min(1, t) = t$.

- **Case 3** ($1 \leq t < 2$): $h(t-\tau)$ completely covers $x(\tau)$. We integrate over all of x: from 0 to 1.
- **Case 4** ($2 \leq t < 3$): The left edge of $h(t-\tau)$ is at $\tau = t-2$, which is inside $[0,1]$. The right edge is at $\tau = t > 1$. So we integrate from $\max(0, t-2) = t-2$ to $\min(1, t) = 1$.

Key Insight

Quick Check: At the boundary between cases, the limits should give the same value. For example, at $t = 1$:

- From Case 2: limits are $[0,1]$, giving $\int_0^1 = 1$
- From Case 3: limits are $[0,1]$, giving $\int_0^1 = 1$ ✓

This continuity check helps verify your cases are correct!

7.6 Worked Examples: Computing the Convolution Integral

7.6.1 Example: Two Rectangular Pulses

Problem: Let $x(t) = u(t) - u(t-1)$ and $h(t) = u(t) - u(t-2)$. Find $y(t) = x(t) * h(t)$.

Step 1: Sketch the signals

- $x(t)$: a pulse of height 1 from $t = 0$ to $t = 1$
- $h(t)$: a pulse of height 1 from $t = 0$ to $t = 2$

Step 2: Flip and shift h

- $h(-\tau)$: pulse from $\tau = -2$ to $\tau = 0$
- $h(t-\tau)$: pulse from $\tau = t-2$ to $\tau = t$

Step 3: Find regions of overlap
We need to identify where $x(\tau)$ and $h(t-\tau)$ are *both* nonzero.

- $x(\tau) \neq 0$ for $0 \leq \tau \leq 1$
- $h(t-\tau) \neq 0$ for $t-2 \leq \tau \leq t$

Case 1: $t < 0$ (no overlap)

$$y(t) = 0$$

Case 2: $0 \leq t < 1$ (partial overlap, h entering from left)

$$y(t) = \int_0^t 1 \cdot 1 \, d\tau = t$$

Case 3: $1 \leq t < 2$ (full overlap with x)

$$y(t) = \int_0^1 1 \cdot 1 \, d\tau = 1$$

Case 4: $2 \leq t < 3$ (partial overlap, h exiting to right)

$$y(t) = \int_{t-2}^1 1 \cdot 1 \, d\tau = 1 - (t-2) = 3 - t$$

Case 5: $t \geq 3$ (no overlap)

$$y(t) = 0$$

Final Answer:

$$y(t) = \begin{cases} 0, & t < 0 \\ t, & 0 \leq t < 1 \\ 1, & 1 \leq t < 2 \\ 3 - t, & 2 \leq t < 3 \\ 0, & t \geq 3 \end{cases}$$

Key Insight

Sanity check: The result is a trapezoidal pulse. This makes sense because convolving two rectangular pulses always produces a trapezoidal (or triangular) shape!

7.6.2 Example: Exponential with Unit Step

Problem: Let $x(t) = e^{-at}u(t)$ and $h(t) = u(t)$, where $a > 0$. Find $y(t) = x(t) * h(t)$.

Solution:

For $t < 0$: There's no overlap, so $y(t) = 0$.

For $t \geq 0$:

$$y(t) = \int_0^t e^{-a\tau} \cdot 1 \, d\tau = \int_0^t e^{-a\tau} \, d\tau$$

$$y(t) = \left[-\frac{1}{a} e^{-a\tau} \right]_0^t = -\frac{1}{a} e^{-at} - \left(-\frac{1}{a} e^0 \right) = \frac{1}{a} \left(1 - e^{-at} \right)$$

Final Answer:

$$\boxed{y(t) = \frac{1}{a} \left(1 - e^{-at} \right) u(t)}$$

Physical interpretation: If $h(t) = u(t)$ represents an integrator and $x(t)$ is a decaying exponential starting at $t = 0$, then $y(t)$ is the running integral of that exponential.

7.7 Properties of LTI Systems

Why This Matters

These properties are not just abstract math—they have practical implications! They tell us how to:

- Simplify calculations (commutativity)
- Analyze series connections of systems (associativity)
- Analyze parallel connections of systems (distributivity)

7.7.1 Commutativity

$$\boxed{x(t) * h(t) = h(t) * x(t)}$$

Meaning: You can interchange the roles of input and impulse response. Flip whichever signal is simpler!

7.7.2 Distributivity

$$\boxed{x(t) * [h_1(t) + h_2(t)] = x(t) * h_1(t) + x(t) * h_2(t)}$$

Application: For **parallel systems**, the equivalent impulse response is:

$$h_{\text{eq}}(t) = h_1(t) + h_2(t)$$

7.7.3 Associativity

$$\boxed{x(t) * [h_1(t) * h_2(t)] = [x(t) * h_1(t)] * h_2(t)}$$

Application: For **cascade (series) systems**, the equivalent impulse response is:

$$h_{\text{eq}}(t) = h_1(t) * h_2(t)$$

Key Insight

Cascade systems are interchangeable! Due to commutativity and associativity, the order of LTI systems in a cascade doesn't affect the overall input-output relationship.

7.7.4 Convolution with the Impulse

$$\boxed{x(t) * \delta(t) = x(t)}$$

The impulse is the **identity element** for convolution (like 1 for multiplication).

7.7.5 Convolution with a Shifted Impulse (Time Delay)

$$\boxed{x(t) * \delta(t - t_0) = x(t - t_0)}$$

Convolving with a shifted impulse **delays** the signal by t_0.

7.8 LTI System Properties from the Impulse Response

Why This Matters

One of the most powerful aspects of LTI systems: you can determine important properties just by looking at $h(t)$!

7.8.1 Memory

Memoryless: The output depends only on the current input value.

An LTI system is memoryless if and only if:

$$\boxed{h(t) = K\delta(t)} \quad \text{for some constant } K$$

In this case: $y(t) = K \cdot x(t)$ (just scaling)

Has memory: If $h(t)$ is anything other than a scaled impulse, the system has memory.

Example:

- $h(t) = 2\delta(t) \Rightarrow$ Memoryless (output $= 2x(t)$)
- $h(t) = e^{-t}u(t) \Rightarrow$ Has memory (output depends on past inputs)

7.8.2 Causality

Causal: The output at time t depends only on the input at time t and earlier (not the future).

An LTI system is causal if and only if:

$$\boxed{h(t) = 0 \quad \text{for } t < 0}$$

Physical interpretation: The system cannot respond *before* the impulse is applied.

Examples:

- $h(t) = e^{-2t}u(t) \Rightarrow$ Causal (zero for $t < 0$)
- $h(t) = e^{2t}u(-t) \Rightarrow$ Non-causal (nonzero for $t < 0$)
- $h(t) = e^{-|t|} \Rightarrow$ Non-causal (nonzero for all t)

Key Insight

For a **causal** system with **causal** input (i.e., $x(t) = 0$ for $t < 0$), the convolution integral simplifies to:

$$y(t) = \int_0^t x(\tau)h(t-\tau)\,d\tau$$

This is often easier to compute!

7.8.3 Stability (BIBO)

BIBO Stable: Bounded Input $\Rightarrow$ Bounded Output

An LTI system is BIBO stable if and only if the impulse response is **absolutely integrable**:

$$\boxed{\int_{-\infty}^{\infty} |h(t)|\,dt < \infty}$$

Intuition: If the "total area" under $|h(t)|$ is finite, then a bounded input cannot produce an unbounded output.

Examples:

- $h(t) = e^{-2t}u(t)$: $\displaystyle\int_0^{\infty} e^{-2t}\,dt = \frac{1}{2} < \infty \Rightarrow$ **Stable**
- $h(t) = e^{t}u(-t)$: $\displaystyle\int_{-\infty}^{0} e^{t}\,dt = 1 < \infty \Rightarrow$ **Stable**
- $h(t) = e^{2t}u(t)$: $\displaystyle\int_0^{\infty} e^{2t}\,dt = \infty \Rightarrow$ **Unstable**
- $h(t) = u(t)$: $\displaystyle\int_0^{\infty} 1\,dt = \infty \Rightarrow$ **Unstable**

Common Mistake

A system can be causal but unstable (like $h(t) = e^{2t}u(t)$), or stable but non-causal (like $h(t) = e^{t}u(-t)$). Causality and stability are independent properties!

7.8.4 Invertibility

An LTI system with impulse response $h(t)$ is **invertible** if there exists an inverse system $h_i(t)$ such that:

$$\boxed{h(t) * h_i(t) = \delta(t)}$$

The cascade of the system and its inverse gives back the original signal.

Example: Time delay and time advance

- If $h(t) = \delta(t - t_0)$ (delay by t_0)

- Then $h_i(t) = \delta(t + t_0)$ (advance by t_0)
- Because: $\delta(t - t_0) * \delta(t + t_0) = \delta(t)$

7.9 The Unit Step Response

The **unit step response** $s(t)$ is the output when the input is the unit step $u(t)$:

$$\boxed{s(t) = u(t) * h(t) = \int_{-\infty}^{t} h(\tau)\, d\tau}$$

Why this formula? Since $u(t - \tau) = 1$ for $\tau \leq t$ and 0 for $\tau > t$:

$$s(t) = \int_{-\infty}^{\infty} h(\tau)u(t - \tau)\, d\tau = \int_{-\infty}^{t} h(\tau)\, d\tau$$

7.9.1 Relationship Between $h(t)$ and $s(t)$

The step response is the integral of the impulse response:

$$s(t) = \int_{-\infty}^{t} h(\tau)\, d\tau$$

The impulse response is the derivative of the step response:

$$\boxed{h(t) = \frac{ds(t)}{dt}}$$

Key Insight

If measuring the impulse response directly is difficult (an ideal impulse is hard to generate physically), you can:

1. Apply a step input $u(t)$
2. Measure the step response $s(t)$
3. Differentiate to get $h(t) = \frac{ds(t)}{dt}$

Example: If $h(t) = e^{-t}u(t)$, find the step response.

$$s(t) = \int_{-\infty}^{t} e^{-\tau}u(\tau)\, d\tau = \int_{0}^{t} e^{-\tau}\, d\tau = 1 - e^{-t} \quad \text{for } t \geq 0$$

So $s(t) = (1 - e^{-t})u(t)$.

7.10 Summary and Key Formulas

Concept	**Formula**
Sifting property	$\int_{-\infty}^{\infty} x(\tau)\delta(\tau - t_0)\, d\tau = x(t_0)$
Signal decomposition	$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t - \tau)\, d\tau$
Impulse response	$h(t) = T\{\delta(t)\}$
Convolution integral	$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)\, d\tau$
Commutativity	$x(t) * h(t) = h(t) * x(t)$
Distributivity	$x * (h_1 + h_2) = x * h_1 + x * h_2$
Associativity	$(x * h_1) * h_2 = x * (h_1 * h_2)$
Identity	$x(t) * \delta(t) = x(t)$
Delay	$x(t) * \delta(t - t_0) = x(t - t_0)$
Memoryless	$h(t) = K\delta(t)$
Causal	$h(t) = 0$ for $t < 0$
BIBO Stable	$\int_{-\infty}^{\infty} \lvert h(t)\rvert\, dt < \infty$
Step response	$s(t) = \int_{-\infty}^{t} h(\tau)\, d\tau$

7.11 Common Mistakes to Avoid

1. **Forgetting to flip before shifting:** In $h(t - \tau)$, you must flip $h(\tau)$ first, then shift. This is the most common error!

2. **Wrong integration limits:** The limits depend on where both signals are nonzero. Always sketch the signals and identify the overlap region.

3. **Confusing t and τ:**

   - t is the output time (the variable in your answer)
   - τ is the dummy integration variable

4. **Forgetting delta sifting rules:** When the delta is outside the integration limits, the integral equals zero.

5. **Confusing stability and causality:** These are independent properties! A system can be stable but non-causal, or causal but unstable.

6. **Computing $\int |h(t)|dt$ without absolute values:** For stability, you need the integral of $|h(t)|$, not $h(t)$.

Key Insight

Study strategy: Practice computing convolutions by hand. Work through the graphical method (flip-shift-multiply-integrate) until it becomes automatic. This builds intuition that will help you throughout the course!

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